

Food for thought: the effect of counterfactual thinking on the use of nutrition information

Khaled Aboulnasr^{1*} and Anu Sivaraman²

¹Lutgert College of Business, Florida Gulf Coast University, Fort Myers, FL 33965, USA

²Alfred Lerner College of Business, University of Delaware, Newark DE 19716, USA

In three experiments, we examine the efficacy of counterfactual thinking (CFT) as a strategy to enhance consumers' motivation to process and use nutrition information on food packages. In the first study, we test whether CFT leads to greater motivation to process nutrition label information in the process of forming product attitudes. We also test whether motivation mediates the relationship between CFT and the influence of the nutrition label in product evaluation. In a second study, we test the effect of upward versus downward CFT on motivation. We also examine whether nutrition information on food packages moderates the relationship between CFT and product attitudes. In a third study, we examined the duration of the motivational effect of CFT. Results from the three studies support the role of CFT as a mechanism that enhances consumers' motivation to elaborate on and use nutrition information to form product attitudes. Copyright © 2010 John Wiley & Sons, Ltd.

Introduction

Aiming at reducing the long-term social and healthcare costs of obesity, Food and Drug Administration (FDA) regulations stemming from the Nutrition Labeling and Education Act (NLEA) required the presence of the standardized nutrition facts panel and regulated health claims on food packages (Keller *et al.*, 1997). While these changes were intended to make nutrition information more comprehensible to consumers, research has suggested that many

consumers do not comprehend nor utilize nutrition information when making consumption decisions (Wansink and Cheney, 2005). While some consumers view nutrition information as misleading or unclear (Szykman *et al.*, 1997), others may lack the motivation to process nutrition information (Moorman, 1996; Keller *et al.*, 1997). Motivation to process nutrition information has been identified as a key determinant to the success of FDA regulations and calls were made to improve the effectiveness of labeling intervention programs by increasing consumers' motivation to utilize nutrition labels (Drichoutis *et al.*, 2006).

The objective of this paper is to examine the efficacy of counterfactual thinking (CFT) as an intervention mechanism to stimulate

*Correspondence to: Khaled Aboulnasr, Lutgert College of Business, Florida Gulf Coast University, 10501 FGCU Blvd. S. Fort Myers, FL 33965, USA.
E-mail: kaboulna@fgcu.edu

consumers' motivation to consider and process nutrition information on food packages. CFT refers to the practice of looking back at past events and mentally imagining how these events could have turned out differently (Page and Colby, 2003). Individuals involved in CFT generate alternative realities to past events (Roese and Olson, 1993); this process has been shown to exert an impact over the processing of subsequently encountered information through a motivational priming process (Nan, 2008). Specifically CFT was found to induce careful scrutiny of subsequently encountered information (Krishnamurthy and Sivaraman, 2002). In the present research, through three experiments, we show that CFT increases the motivation to process nutrition information. This motivation is translated into greater usage and more careful examination of both health claims and nutrition information on food products.

Conceptual background and hypotheses

Motivation

We define motivation as "consumers' goal-directed arousal to process nutrition information" (Balasubramanian and Cole, 2002). Motivation is an important catalyst in most nutrition information search and processing models. It has been found to be an important moderator of the effects of nutrition labels on dietary choices and consumption behavior and has also been identified as an important predictor of information acquisition (Keller *et al.*, 1997). Moorman (1990) found that motivation influenced how consumers processed nutrition information and made nutrition choices.

The elaboration likelihood model lends support to the role of motivation in processing nutrition information. More motivated consumers are more likely to expend greater cognitive effort to elaborate on diagnostic information viewed as central to the evaluation process such as the nutrition facts panel (Keller *et al.*, 1997). Hence, encouraging

active elaboration of nutrition information through increasing consumers' motivation to process is critical in helping consumers make more informed food choices.

Counterfactual thinking

The fundamental feature of CFT is the generation of alternative realities in which a factual outcome is undone through mentally altering one or more antecedent event(s) and then assessing the consequence of the alteration (Roese, 1997). Similar to the negativity bias, counterfactual thoughts are more likely to be generated in response to negative than to positive events (Roese, 1997). Negative events were shown to generate more cognitive simulation as a means to understand the causes for such events and the actions that could have been taken to prevent them (Page and Colby, 2003). For example, if Chris learns that his cholesterol level is high because of inferior dietary choices and that his health is at risk, he may think that he could have avoided such risk if only he had made healthier nutrition decisions.

Wells and Gavanski (1989) showed that CFT serves a preparative function by affecting judgments of causality. The mental simulation involved in generating alternative realities in which a factual outcome's antecedents are cognitively mutated and the impact of such alteration on the outcome is observed, may shape future actions by enhancing the intention to perform success-facilitating behaviors (Page and Colby, 2003). Consequently, CFT may be viewed as a corrective mechanism that helps establish the causes of events and hence, to the extent that a similar situation may take place in the future, it may help consumers take actions intended to improve future outcomes (Roese, 1997). More recently, Krishnamurthy and Sivaraman (2002) suggested that the preparative function of CFT may also include an information processing consequence that influences future problem solving behavior. Drawing on the Elaboration Likelihood Model, the authors demonstrated that CFT may be

viewed as a problem-solving process in which people arrive at how an outcome could have turned out differently by making distinctions between good and bad solutions to the problem. They proposed that engaging (versus not engaging) in CFT leads to a greater sensitivity to the strength of the argument in advertising messages.

In the present research, we argue that motivation to process information will drive the effect of CFT on the processing of subsequently encountered nutrition information. We expect consumers who engage in CFT in response to a negative diet-related health condition will identify ways in which that negative outcome could have been avoided. This process, coupled with judgments of causality, will increase consumers' motivation to avoid similar negative outcomes in the future. The search and alignment model in judgment and decision making proposed by Pham and Muthukrishnan (2002) may also explain the proposed effect. When consumers are exposed to a negative diet-related experience and then asked to generate counterfactual thoughts of how it could have been avoided, this process triggers a search for alternative routes in which the negative outcome is compared to a more positive one. This complex progression may lead to a revision of existing practices and a greater motivation to diagnostically examine subsequently encountered information.

In summary, we argue that in response to a negative diet-related event, consumers who engage in CFT (e.g., if only thoughts) will be more motivated to scrutinize subsequently encountered nutrition information on food packages, and allocate more weight to the role of nutrition information in overall product evaluation. Hence, we put forth the following hypothesis:

H1: Following exposure to a negative diet-related event, consumers who engage in counterfactual thinking will report greater motivation to process nutrition information compared to those who do not.

Having greater motivation to process nutrition information, we expect consumers who undergo CFT to report a greater influence of the nutrition label in their product evaluation. While other sources of information may be available on a food package such as brand name or a promotional claim, we expect consumers who generate counterfactual thoughts to employ more careful processing of information and hence, give more weight to the nutrition label compared to consumers who do not generate such thoughts. This argument finds support in the elaboration likelihood model. More motivated consumers are more likely to expend effort to elaborate on diagnostic information viewed as central to the evaluation process such as the nutrition facts panel. Hence, we posit the following:

H2: Following exposure to a negative diet-related event, consumers who engage in counterfactual thinking will report greater influence of the food product's nutrition label in overall product evaluation compared to those who do not.

If in fact the effect of CFT on consumers' processing of nutrition information and use of the nutrition label for judgment is routed through motivation, then we should expect a strong mediating effect of motivation on the relationship between CFT and the influence of the nutrition panel. Hence, we hypothesize that:

H3: Motivation to process nutrition information mediates the effect of counterfactual thinking on consumers' self-reported influence of the nutrition label in overall product evaluation.

A number of research studies examined whether the presence of health claims on food packages interfered with consumers' motivation and ability to process information in the nutrition panel (Kozup *et al.*, 2003). While Ford *et al.* (1996) suggested that consumers treat health claims and the nutrition label as

independent sources of information, Roe *et al.* (1999) asserted that for some consumers, health claims may be used as a substitute for the nutrition facts panel. Hence, in our first experiment, we control for the potential effect of a health claim by examining the effect of CFT both in the presence and absence of health claims.

In Experiment 1, we tested the effects of engaging (versus not engaging) in CFT on participants' motivation (H1) and on the influence of the nutrition information (H2). We also examined whether motivation mediates the effect of CFT on the influence of nutrition information in product evaluation (H3).

Experiment 1

Design and Stimuli

The study was a $2(\text{CFT vs. no CFT}) \times 2(\text{health claim vs. no health claim})$ between subjects' design. This was followed by a test of mediation to test H3. A packaged frozen food product was the focus of the study. A mock of the front and back panels of the package were created similar to the stimuli used by Keller *et al.* (1997). We used the fictitious brand name "Blue Ribbon" which was used by Kozup *et al.* (2003). The nutrition panel on the back of the mock package contained relatively favorable nutrition information (see Appendix A).

Participants and procedure

Sixty seven undergraduate students at a south-eastern public university participated in the study in return for extra credit. They were randomly assigned to one of the four conditions. The study was divided into two stages. Participants were not informed of an impending second stage. All study participants were first exposed to a hypothetical episode that described the lifestyle of Chris, a 26-year old college student who has led a relatively unhealthy life style in terms of diet and exercise and has recently been diagnosed with severe constipation, described as a serious health

condition. Following exposure to this episode, CFT was manipulated as described below. Following this, all participants completed the second stage of the experiment in which they were exposed to the front and back panels of the mock food package. Participants then responded to rating scales pertaining to the dependent variables.

Independent variables

Counterfactual thinking

CFT manipulation was conducted in a manner similar to prior research (Page and Colby, 2003). After reading the hypothetical scenario, participants in the CFT condition were instructed to write down their "If only" thoughts of actions that in retrospect the character in the scenario could have done to avoid his negative health condition. Respondents in the no CFT condition only read the hypothetical scenario but were not instructed to generate counterfactual thoughts.

Health claims

Similar to Ford *et al.* (1996), the health claim version made the explicit claim "does your heart good," "low in cholesterol," and "low in calories." It also showed a picture of a heart and stated the amount of cholesterol. In contrast, the promotional claim version made the claims, "ultimate in cooking experience," "tastes great," and "ready in minutes" and showed a picture of a clock (see Appendix B).

Dependent variables

Whenever multiple items were used, the mean rating of these items was used as a measure of the construct.

Motivation to read the nutrition label

Study participants rated their motivation to read the nutrition label (Cronbach's $\alpha = .90$) on a three-item seven-point scale (I was

motivated/interested/I cared – to read the nutrition information on the nutrition label).

and health claim was observed. Follow up univariate results are detailed below.

Influence of the nutrition label

The extent to which the nutrition label influenced the evaluation of the product was measured using a two-item seven-point rating scale (Cronbach's $\alpha = .91$). Respondents were asked to answer: To what extent did the information from the nutrition label influence your evaluation of the Blue Ribbon product; the items were anchored by “did not influence my evaluation at all” and “Influenced my evaluation a lot” and “Was not important at all in my evaluation” and “Was very important in my evaluation”.

Results

Manipulation checks

The generation of counterfactual thoughts is known to be associated with greater negative feelings as a result of the comparison between the actual negative outcome and an imagined more positive outcome. Respondents were asked to rate the degree to which they think the character in the hypothetical scenario felt bad for his situation. Subjects in the CFT condition associated greater levels of negative emotion with the character compared to participants in the no CFT condition (5.50 vs. 4.86; $t(65) = 2.46, p < .05$).

The health claim manipulation was checked by comparing respondents' beliefs in the heart healthiness of the product. Respondents in the health claim condition gave the product a more favorable heart rating than did respondents in the no health claim condition (4.72 vs. 3.88; $t(65) = -2.57, p < .02$).

Multivariate results

MANOVA results revealed a significant main effect of CFT (Wilk's $\lambda = 0.90, F = 3.39, p < 0.05$) on the dependent variables. No significant interaction effect between CFT

Motivation to read the nutrition label

In support of H1, univariate test results revealed a significant main effect for CFT, ($F(1, 63) = 4.75, p < .05$). Those who engaged in CFT reported higher motivation to process nutrition information than those who did not (5.57 vs. 4.77). Presence of a health claim had no significant effect on motivation. Furthermore, no significant interaction effects were observed between CFT and health claim on motivation.

Influence of nutrition label in product evaluation

Results revealed a significant main effect of CFT on self-reported influence of the nutrition panel in product evaluation ($F(1, 63) = 5.94, p < .02$). The CFT condition reported greater influence of the nutrition label in product evaluation compared to the no CFT condition (5.91 vs. 5.02). No significant main or interaction effects were observed for health claims on the influence of the nutrition label. This result provides support for H2. Multivariate and univariate results are presented in **Table 1**. The cell means for all the dependent variables are available in **Table 2**.

Mediating role of motivation

Following the procedure suggested by Baron and Kenny (1986), we tested the mediating role of motivation between CFT and the influence of the nutrition panel in product evaluation. First, using a regression equation we used motivation as the criterion variable and CFT as the predictor variable. We found that CFT increased motivation to process nutrition information ($B = .69, t(65) = 2.15, p < .04$). Second, we regressed influence of the nutrition label on CFT. As expected, CFT increased the influence of the nutrition label in product evaluation. ($B = .75, t(65) = 2.37,$

Table 1. Experiment 1 – multivariate and univariate results

Independent variables	Manova results		Univariate <i>F</i> -values	
	Wilks' λ	<i>F</i> -value	Motivation	Use of nutrition label
Main effects				
CFT	0.90	3.39*	4.75*	5.94*
HC	0.96	1.31	2.66	0.71
Interaction effects				
CFT $\times$ HC	0.97	0.85	1.69	0.81

CFT = counterfactual thinking; HC = health claim.

* $p < .05$.

$p < .03$). Third, we regressed self-reported influence of the nutrition label on motivation. Motivation had a significant effect on the influence of the nutrition label ($B = .62$, $t(65) = 6.46$, $p < .001$). Last, we used self-reported influence of the nutrition label as the criterion variable and CFT and motivation as predictors in the same regression equation. The coefficient of CFT became smaller and statistically insignificant ($B = .34$, $t(65) = 1.29$, $p = .20$) whereas the influence of motivation was significant ($B = .59$, $t(65) = 5.94$, $p < .001$). A Sobel (1982) test indicated that the mediation effect was statistically significant ($Z = 2.02$, $p < .05$). Overall, these results

indicate that motivation to process nutrition information mediates the relationship between CFT and the influence of the nutrition label and provide support for H3.

Discussion

Results from Experiment 1 support the motivational outcomes of CFT on the processing of subsequently encountered information. Results support the argument that engaging in CFT leads to greater motivation to process nutrition information on food packages (H1). CFT was also shown to increase the influence of the nutrition facts panel in product evaluation (H2). Furthermore, results demonstrated that motivation mediates the relationship between CFT and the role of the nutrition label in consumers' product evaluation (H3). In contrast to Roe *et al.* (1999), results showed that the presence or absence of a health claim on the front of the package did not have a main or interaction effect on the motivation to utilize the nutrition facts panel.

While Experiment 1 has offered support for H1, H2, and H3 establishing the effect of CFT on motivation and usage of nutrition information, it had some limitations. First, it did not examine the effect of CFT in light of the direction of comparison. Prior research had indicated that the direction of the counterfactual thoughts may influence the behavioral outcomes of such thoughts. Second, Experiment 1

Table 2. Experiment 1 – dependent variable means

	Motivation ^a	Use of nutrition label ^a
Main effects		
CFT	5.57	5.91
No CFT	4.77	5.02
HC	5.45	5.61
No HC	4.90	5.32
Interaction effects		
CFT		
HC	5.63	5.90
No HC	5.51	5.92
No CFT		
HC	5.27	5.31
No HC	4.28	4.72

CFT = Counterfactual thinking; HC = Health claim.

^aThese measures are based on seven-point scales; higher scores indicate more greater motivation/use of nutrition label.

did not address how CFT influenced product attitudes and evaluation. Third, we did not test how nutrition information on the food package interacted with CFT to shape product attitudes. As such, in study 2 we address these concerns.

Directional effect of counterfactual thinking

The preparative and motivational functions of CFT are best viewed in light of the direction of comparison. Counterfactual thoughts may generate alternatives that are better than reality through upward counterfactuals (i.e., if only) or worse than reality through downward counterfactuals (i.e., well at least). While downward counterfactuals have been known to generate positive emotions such as relief and respite, upward counterfactuals have been found to generate negative emotions, particularly regret and disappointment (Page and Colby, 2003). Because of the desire to prevent such negative emotions coupled with negative outcomes in the future, upward compared to downward CFT was found to be more useful in prescribing success-facilitating future behavior (Page and Colby, 2003). Hence, upward CFT is more corrective in nature and compared to downward CFT is more potent in identifying causally effective antecedents that may then be used to enhance future performance (Roese, 1997). Following exposure to a negative diet-related event, we expect upward CFT to generate greater motivation to process subsequently encountered nutrition information in order to prevent a similarly negative outcome in the future compared to downward CFT. Therefore,

H4: Following exposure to a negative diet-related event, consumers who engage in upward counterfactual thinking will report greater motivation to process nutrition information compared to those who engage in downward counterfactual thinking.

Similar to the nutrition panel, health and nutrition claims (i.e., does your heart good)

placed on the front of a food package are considered a source of evaluative information (Ford *et al.*, 1996). They alert consumers to a food product's potential health benefits or key nutrient information. Such claims represent an easily accessible and independent source of product-related information that may be used in building consumers' product attitudes and evaluations (Ford *et al.*, 1996).

Given that health claims represent an independent source for evaluative judgment, and building on the argument that upward CFT creates greater elaboration of subsequently encountered information, we should then expect product attitudes of consumers who generate upward counterfactuals to be more sensitive to the presence or absence of health claims on a food package. We thus predict that compared to food packages with no health claims, the presence of a health claim generates more favorable product attitudes for consumers who engage in upward CFT than for consumers who generate downward counterfactuals. Hence,

H5: Health claims moderate the effect of counterfactual thinking on product attitudes. Compared to food packages on which no health claims are made, packages with health claims lead to a more favorable product evaluation for consumers who engage in upward counterfactual thinking than those who engage in downward counterfactual thinking.

Consumers who have higher levels of motivation have been shown to have a greater disposition towards acquiring and scrutinizing nutrition information on food products (Moorman, 1996). Furthermore, more motivated consumers place more emphasis on more diagnostic information and they perceive the specific information contained in the nutrition label as essential for product judgment (Keller *et al.*, 1997). These findings are supported by the results of our first experiment illustrating that motivation mediates the relationship between CFT and the usage of the nutrition label. The elaboration likelihood model pro-

vides further support for the information processing outcomes of motivation. In the case of a food product, more motivated consumers are more likely to elaborate on diagnostic information such as the nutrition label perceived as important to the judgment task (Keller *et al.*, 1997).

Through the first experiment, we demonstrated that CFT leads to greater motivation to process nutrition information and consequently greater usage and scrutiny of such information. Hence, we should expect those who engage in upward CFT to be more responsive to the favorability of the product's nutrition label information. Compared to downward CFT, upward CFT should lead to a more favorable attitude towards the product when the nutrition label contains favorable information. Similarly, when the nutrition label contains unfavorable nutrition information, given the more careful processing and the greater emphasis placed on the nutrition label, we should also expect consumers who generate upward counterfactuals to have less favorable product attitudes compared to those who engage in downward CFT. In light of the preceding argument, we posit the following:

H6: The valence of the nutrition label moderates the effect of counterfactual thinking on product attitudes. (a) Food packages that have a favorable nutrition label will lead to more favorable product attitudes for consumers who engage in upward versus downward counterfactual thinking and (b) food packages that have an unfavorable nutrition label will lead to less favorable product attitudes for consumers who engage in upward versus downward counterfactual thinking.

In Experiment 2 we tested H4, H5, H6a, and H6b. We compared between the effects of upward versus downward CFT on motivation and tested whether nutrition information (i.e., nutrition label and health claim) moderates the relationship between CFT and product attitudes.

Experiment 2

Design and stimuli

This study was a $2 \times 2 \times 2$ between subjects design using CFT (*upward vs. downward*), health claim (*present vs. absent*) and nutrition panel (favorable vs. unfavorable). The same packaged food product used in Experiment 1 was utilized for the second experiment.

Participants and procedure

Two hundred and three undergraduate students at a southeastern public university participated in the study in return for extra credit. Participants were randomly assigned to one of eight conditions. Similar to Experiment 1, in the first stage of the study, CFT was manipulated. In the second stage health claim and nutrition panel were manipulated and dependent measures were captured.

Independent variables

Counterfactual thinking

After reading the scenario, respondents in the upward CFT condition were asked to list their "if only" thoughts while participants in the downward CFT condition were asked to list their "well at least thoughts".

Health claims and nutrition panel

Health claims were manipulated in the same manner as in Experiment 1. The nutrition panel was manipulated by altering the nutrient content levels in the panel (see Appendix A).

Dependent variables

Motivation to process nutrition information

Motivation was measured through the same items used in Experiment 1 (Cronbach's $\alpha = .92$).

Product attitudes

Product attitudes were assessed using three, seven-point items anchored by unfavorable/

favorable, bad/good, and negative/positive (Cronbach's $\alpha = .96$).

Results

Manipulation checks

Upward versus downward CFT manipulation was checked using a qualitative review of the thoughts generated. The review revealed that participants generated thoughts in accordance with the instructed direction. Furthermore, as was expected, subjects in the upward CFT condition reported greater levels of regret compared to participants in the downward CFT condition (5.61 vs. 5.13; $t(201) = 2.81, p < .01$).

The health claim manipulation was successful as respondents in the health claim present group gave the product a more favorable heart rating than did the health claim absent group (3.99 vs. 3.34; $t(201) = -3.13, p < .01$). Nutrition panel favorability manipulation was checked using a one-item seven-point scale ("I think the nutrition level of this product is ..." with endpoints of poor and good). Respondents in the favorable nutrition label condition rated the product significantly higher in nutrition level compared to respondents in the unfavorable nutrition label condition (4.31 vs. 2.96; $t(201) = -8.22, p < .001$).

Multivariate results

MANOVA results revealed a significant main effect of CFT (Wilk's $\lambda = 0.95, F = 4.99, p < 0.01$), a significant interaction effect of CFT and favorability of nutrition label (Wilk's $\lambda = 0.96, F = 4.35, p < .001$) and a significant interaction effect of CFT and health claim (Wilk's $\lambda = 0.91, F = 9.00, p < .001$). Follow up univariate results are detailed below. No significant three way interactions or health claim-nutrition panel favorability interactions on product attitudes were observed.

Motivation to process nutrition information

The effect of upward versus downward CFT on the motivation to process nutrition information

was examined. Results revealed a significant main effect of CFT on motivation, ($F(1, 195) = 9.57, p < .01$). Those who engaged in upward CFT reported higher motivation to process nutrition information ($M = 5.51$) compared to those who engaged in downward CFT ($M = 4.97$). This result supports H4.

Overall product attitudes

H5 predicted that health claims moderate the effects of CFT on product attitude. Consistent with H5, results show a significant interaction effect between CFT and health claims ($F(1, 195) = 17.16, p < .001$). In the health claim present condition, respondents who generated upward counterfactuals rated the product more favorably ($M = 4.13$) compared to those who generated downward counterfactuals ($M = 3.51$). In the health claim absent condition, respondents who generated upward counterfactuals rated the product less favorably ($M = 3.50$) compared to those who generated downward counterfactuals ($M = 4.43$). These results support H5.

H6a and H6b predicted that the favorability of the nutrition label moderated the effect of CFT on product evaluation. Consistent with H6a and H6b, results revealed a significant interaction effect between CFT and the favorability of the nutrition panel on product evaluation $F(1, 195) = 7.63, p < .01$. In the favorable nutrition label condition, respondents who generated upward counterfactuals evaluated the product more favorably ($M = 4.78$) compared to respondents who generated downward counterfactuals ($M = 4.42$). In the unfavorable nutrition label condition, respondents who created upward counterfactuals rated the product less favorably ($M = 2.84$) compared to respondents in the downward counterfactual condition ($M = 3.52$). These results support H6a and H6b. Multivariate and univariate results are presented in Table 3. Cell means for dependent variables are available in Table 4.

Discussion

Results successfully demonstrated that inducing upward CFT prior to the exposure to a

Table 3. Experiment 2 – multivariate and univariate results

Independent variables	Manova results		Univariate <i>F</i> -values	
	Wilks' λ	<i>F</i> -value	Motivation	Product attitude
Main effects				
CFT	0.95	4.99**	9.57**	0.70
HC	1.00	0.30	0.03	0.57
NL	0.77	28.98*	3.07	56.39*
Interaction effects				
CFT $\times$ HC	0.91	9.00*	0.58	17.16*
CFT $\times$ NL	0.96	4.35*	0.83	7.63**

CFT = counterfactual thinking; NL = nutrition label; HC = health claim.

* $p < .001$;

** $p < .01$.

food product has a significant effect on the motivation to process nutrition information. As evidence of such greater motivation, results indicate that consumers in the upward CFT were better able to discriminate between favorable and unfavorable nutrition infor-

mation as reflected in their product attitudes compared to those in the downward CFT condition. Those who generated upward counterfactual thoughts had significantly higher product attitudes in the presence of health claims or when the nutrition label was favorable but lower evaluation in the absence of a health claim or when the nutrition label was unfavorable compared to those who generated downward counterfactuals. While we were able to demonstrate that CFT leads to more careful processing of nutrition information on food packages, in both experiments the motivational effects were captured immediately following the CFT task. However in a more realistic setting, CFT may not immediately precede nutrition label processing or product evaluation. Hence, a length of time may exist between generating counterfactuals and nutrition label processing that typically occurs at the point of purchase. As such, in a third experiment, we take a longitudinal approach and test the durational effects of CFT. We investigate whether the motivational effect will last 7 days after the CFT task is completed.

Table 4. Experiment 2 – dependent variable means

	Motivation ^a	Product attitude ^a
Main effects		
Upward CFT	5.51	3.81
Downward CFT	4.97	3.97
HC	5.22	3.82
No HC	5.25	3.96
Favorable NL	5.08	4.60
Unfavorable NL	5.39	3.18
Interaction effects		
Upward CFT		
HC	5.56	4.13
No HC	5.46	3.50
Downward CFT		
HC	4.89	3.51
No HC	5.05	4.43
Upward CFT		
Favorable NL	5.43	4.78
Unfavorable NL	5.58	2.84
Downward CFT		
Favorable NL	4.73	4.42
Unfavorable NL	5.20	3.52

CFT = counterfactual thinking; HC = health claim; NL = nutrition label.

^aThese measures are based on seven-point scales; higher scores indicate higher motivation/more favorable product attitude.

Experiment 3

Design

This study employed a 2×2 between subjects design using CFT (*upward vs. downward*),

and nutrition panel (favorable vs. unfavorable). The same food product that was used in experiments 1 and 2 was utilized.

Participants, procedure, and measures

Fifty undergraduate students at a southeastern public university participated in the study in return for extra credit. The experiment was conducted in a manner similar to the first two experiments except that the second stage of the study was conducted 7 days after the first. Motivation to process nutrition information (Cronbach's $\alpha = .94$) and product attitude (Cronbach's $\alpha = .96$) were measured using the same items as in Experiment 2.

Results

Manipulation checks

Generated counterfactual thoughts were in accordance with instructed direction. Furthermore, subjects in the upward CFT reported greater levels of regret compared to participants in the downward CFT condition (5.46 vs. 4.41; $t(48) = 2.51$, $p < .05$). Nutrition panel favorability manipulation was successful. Participants in the favorable nutrition label condition rated the product significantly higher in nutrition level compared to respondents in the unfavorable nutrition label condition (4.92 vs. 2.77; $t(48) = 5.47$, $p < .001$).

Multivariate results

MANOVA results revealed a significant main effect of CFT (Wilk's $\lambda = 0.56$, $F = 16.9$, $p < .001$), and a significant main effect of the favorability of nutrition label (Wilk's $\lambda = 0.87$, $F = 3.12$, $p = .05$). No significant multivariate interaction effects of CFT and favorability of the nutrition label were observed. Follow up univariate results are detailed below.

Motivation to process nutrition information

Motivation to process nutrition information was examined 7 days after the CFT task was

completed. Results revealed a significant main effect of CFT on motivation ($F(1, 46) = 32.40$, $p < .001$). Those who engaged in upward CFT reported higher motivation to process nutrition information ($M = 6.42$) compared to those who engaged in downward CFT ($M = 4.55$). This result provides support to the longer term motivational effects of CFT.

Overall product attitudes

Consistent with our predictions, results revealed a significant long-term interaction effect between CFT and the favorability of the nutrition panel on product attitude ($F(1, 46) = 3.08$, $p < .10$). In the favorable nutrition label condition, respondents who generated upward counterfactuals evaluated the product more favorably ($M = 4.71$) compared to respondents who generated downward counterfactuals ($M = 3.66$). In the unfavorable nutrition label condition, respondents who created upward counterfactuals rated the product less favorably ($M = 2.95$) compared to respondents in the downward CFT condition ($M = 3.37$).

These results support the longer term effects of CFT. They provide evidence of the ability of upward counterfactual thinking to lead to greater motivation to process nutrition information and influence product attitudes 7 days after the generation of such thoughts. These results suggest that counterfactual thinking does not have to occur at the point of purchase where the actual processing of the nutrition information takes place in order to influence motivation. Multivariate and univariate results are presented in Table 5. Cell means are available in Table 6.

General discussion

In three studies, we examined the efficacy of CFT as a motivational strategy in the context of nutrition information. Specifically, we examined the effect of generating counterfactual thoughts following the exposure to a negative diet-related health event on the motivation to utilize subsequently encountered nutrition information on food packages. In the first two

Table 5. Experiment 3 – multivariate and univariate results

Independent variables	Manova results		Univariate <i>F</i> -values	
	Wilks' λ	<i>F</i> -value	Motivation	Product attitude
Main effects				
CFT	0.56	16.90*	32.40*	0.58
NL	0.87	3.13**	0.00	6.31**
Interaction effects				
CFT $\times$ NL	0.93	1.72	0.77	3.08**

CFT = counterfactual thinking; NL = nutrition label.

* $p < .001$;** $p < .01$.

experiments, this effect was examined immediately following the CFT task and in the third experiment, 7 days after such task. Taken together, the outcome of the three studies support the role of CFT as a cognitive activity that leads to greater motivation and usage of nutrition information on food products in both the short and long runs. Results indicate that CFT leads to greater motivation to process nutrition information (H1 supported). Similarly, results show that CFT leads to greater influence of nutrition information in product evaluation (H2 supported). We also found that motivation mediates the relationship between

CFT and influence of nutrition information in product evaluation (H3 supported).

When we looked at the direction of comparison, upward CFT was found to lead to greater motivation compared to downward CFT (H4 supported). In line with our prediction, compared to food packages with no health claims, when a health claim was present, respondents in the upward CFT condition had significantly more favorable product attitudes compared to those in the downward CFT condition (H5 supported). Similarly, respondents in the upward CFT condition had significantly less favorable attitudes for products with unfavorable nutrition labels and significantly more favorable attitudes for products with favorable nutrition labels compared to respondents in the downward CFT condition (H6a, H6b supported). Furthermore, we demonstrated that the motivational and attitudinal outcomes of CFT may last up to 7 days following the generation of such thoughts.

The findings from this research have important implications for consumer welfare. The consumption of food products that contain high levels of harmful nutrients has been linked to greater risk of obesity and other diet-related diseases (Keller *et al.*, 1997). The significant difference in product attitudes expressed in response to favorable and unfavorable nutrition labels as well as to the presence of a health claim suggests that upward CFT is effective in enhancing consumers' motivation to carefully elaborate on nutrition information

Table 6. Experiment 3 – cell means

	Motivation ^a	Product attitude ^a
Main effects		
Upward CFT	6.42	3.83
Downward CFT	4.55	3.52
Favorable NL	5.48	4.18
Unfavorable NL	5.49	3.16
Interaction effects		
Upward CFT		
Favorable NL	6.27	4.71
Unfavorable NL	6.57	2.95
Downward CFT		
Favorable NL	4.69	3.66
Unfavorable NL	4.40	3.37

CFT = counterfactual thinking; NL = nutrition label.

^aThese measures are based on seven-point scales; higher scores indicate higher motivation, more favorable product attitude.

and hence making more careful and informed food choices. These findings frame CFT as a valuable public policy instrument that can be used to aid in the success of NLEA labeling changes that require food manufacturers to place clear labels on food products. The present research suggests that using CFT in effectively designed public service announcements may have significant consequences on consumers' motivation. The benefits of such motivation are mirrored in the greater usage and utilization of nutrition information on food packages for product evaluation as well as the significant difference in product attitudes based on their nutritional value.

Limitations and future research

The generalizability of our findings may be restricted due to several limitations. First, data were collected in a classroom setting and therefore, variables such as brand names, displays and other promotions were not examined. Another possible limitation relates to the elements of CFT examined. In the first experiment we only contrasted the effects of engaging versus not engaging in CFT. In the second experiment we contrasted the effects of upward versus downward CFT. Future research may want to examine the effects of other aspects of CFT such as the structure (additive versus subtractive) or the valence (positive versus negative) of CFT. Furthermore, future research may examine other dependent variables, such as consumers' purchase intentions. Despite the study limitations, overall this paper represents an important attempt to use CFT as a strategy to influence consumers' motivation and attitude formation. This study has demonstrated both the importance and the potential of CFT as a potent mechanism in motivating consumers to scrutinize and elaborate on subsequently encountered information.

Biographical notes

Khaled Aboulnasr is an Assistant Professor of Marketing at the Lutgert College of Business, Florida Gulf Coast University. He received his

Ph.D. in Business Administration from the University of Houston. His research interests are in the areas of innovation, competitive response and consumers' response to new products with emphasis on consumers' information processing and decision making. His research has appeared in the *Journal of Marketing* among others.

Anu Sivaraman is an Assistant Professor of Marketing at the Alfred Lerner College of Business, University of Delaware. She received her Ph.D. in Business Administration from the University of Houston. Her research focuses on how counterfactual thinking can be used for understanding and instigating changes in consumption patterns. Her research has appeared in the *Journal of Consumer Research*. She serves as an adhoc reviewer for *Journal of Consumer Psychology* and *International Journal of Forecasting*.

References

- Balasubramanian SK, Cole C. 2002. 'Consumers' search and use of nutrition information: the challenge and promise of the nutrition labeling and education act. *Journal of Marketing* 66(July): 112-127.
- Baron M, Kenny A. 1986. The moderator-mediator variable distinction in social psychological research: conceptual, strategic, and statistical considerations. *Journal of Personality and Social Psychology* 51: 1173-1182.
- Drichoutis AC, Lazaridis P, Nayga RM Jr. 2006. 'Consumers' use of nutritional labels: a review of research studies and issues. *Academy of Marketing Science Review* 9: 1-22.
- Ford GT, Hastak M, Mitra A, Ringold DJ. 1996. Can consumers interpret nutrition information in the presence of a health claim? A laboratory investigation. *Journal of Public Policy and Marketing* 15(1): 16-27.
- Keller SB, Landry M, Olson J, Velliquette AM, Burton S, Andrews JC. 1997. The effects of nutrition package claims, nutrition facts panels, and motivation to process nutrition information on consumer product evaluations. *Journal of Public Policy and Marketing* 16(2): 256-269.
- Kozup JC, Creyer EH, Burton S. 2003. Making healthful food choices: the influence of health

- claims and nutrition information on consumers' evaluations of packaged food products and restaurant menu items. *Journal of Marketing* 67(April): 19–34.
- Krishnamurthy P, Sivaraman A. 2002. Counterfactual thinking and advertising responses. *Journal of Consumer Research* 29(March): 650–658.
- Moorman C. 1990. The effects of stimulus and consumer characteristics on the utilization of nutrition information. *Journal of Consumer Research*, 17(December): 362–374.
- Moorman C. 1996. A quasi-experiment to assess the consumer and informational determinants of nutrition information processing activities: The case of the Nutrition Labeling and Education Act. *Journal of Public Policy and Marketing* 15(1): 28–44.
- Nan X. 2008. The pursuit of self-regulatory goals how counterfactual thinking influences advertising persuasiveness. *Journal of Advertising* 37(1): 17–27.
- Page CM, Colby PM. 2003. If only I hadn't smoked: The impact of counterfactual thinking on a smoking-related behavior. *Psychology and Marketing* 20(11): 955–976.
- Pham MT, Muthukrishnan AV. 2002. Search and alignment in judgment revision: implications for brand positioning. *Journal of Marketing Research* 39: 18–30.
- Roe B, Levy AS, Derby BM. 1999. The impact of health claims on consumer search and product evaluation outcomes: Results from FDA experimental data. *Journal of Public Policy and Marketing* 18(1): 89–105.
- Roese NJ. 1997. Counterfactual thinking. *Psychology Bulletin* 121(1): 133–148.
- Roese NJ, Olson JM. 1993. The structure of counterfactual thought. *Personality and Social Psychology Bulletin* 19: 312–319.
- Sobel ME. 1982. Asymptotic intervals for indirect effects in structural equations models. In *Sociological Methodology 1982*, S Leinhardt (Ed.). Jossey-Bass: San Francisco; 290–312.
- Szykman LR, Bloom PN, Levy AS. 1997. A proposed model of the use of package claims and nutrition labels. *Journal of Public Policy and Marketing* 16(2): 228–241.
- Wansink B, Cheney MM. 2005. Leveraging FDA health claims. *Journal of Consumer Affairs* 39(2): 386–398.
- Wells GL, Gavanski I. 1989. Mental simulation of causality. *Journal of Personality and Social Psychology* 56: 161–169.

Appendix A: Nutrition label on the back of the package**Favorable nutrition panel**

Nutrition Facts			
Serving Size: 1 Package (320g)			
Serving Per Container: 1			
Amount Per Serving			
Calories 170 Calories from Fat 15			
%Daily Value*			
Total Fat 2g			3%
Saturated Fat 0.5g			3%
Cholesterol 20mg			7%
Sodium 410mg			17%
Total Carbohydrate 26g			9%
Dietary Fiber 6g			24%
Sugars 6g			
Protein 11g			
Vitamin A 25% Vitamin C 15%			
Calcium 15% Iron 10%			
*Percent Daily Values are based on a 2,000 calorie diet. Your daily values may be higher or lower depending on your calorie needs.			
	Calories	2000	2500
Total Fat	Less than	65g	80g
Saturated Fat	Less than	20g	25mg
Cholesterol	Less than	300mg	300mg
Sodium	Less than	2400mg	2400mg
Total Carbohydrate		300mg	375mg
Dietary Fiber		25mg	30mg
Calories Per gram:			
Fat 9	Carbohydrates 4		Protein 4

Unfavorable nutrition panel

Nutrition Facts			
Serving Size: 1 Package (320g)			
Serving Per Container: 1			
Amount Per Serving			
Calories 360 Calories from Fat 190			
%Daily Value*			
Total Fat 21g			32%
Saturated Fat 10g			50%
Cholesterol 102mg			34%
Sodium 830mg			35%
Total Carbohydrate 26g			9%
Dietary Fiber 2g			8%
Sugars 5g			
Protein 18g			
Vitamin A 10% Vitamin C 2%			
Calcium 15% Iron 10%			
*Percent Daily Values are based on a 2,000 calorie diet. Your daily values may be higher or lower depending on your calorie needs.			
	Calories	2000	2500
Total Fat	Less than	65g	80g
Saturated Fat	Less than	20g	25mg
Cholesterol	Less than	300mg	300mg
Sodium	Less than	2400mg	2400mg
Total Carbohydrate		300mg	375mg
Dietary Fiber		25mg	30mg
Calories Per gram:			
Fat 9	Carbohydrates 4		Protein 4

Appendix B: Health claim on the front of the package**Health claim present**

BLUE RIBBON

IT DOES YOUR HEART GOOD 

Low in Calories

Low in Cholesterol

Fully Cooked

Keep Frozen

NET WT: 11.25 OZ (320g)

Health claim absent

BLUE RIBBON

The Ultimate Cooking Convenience 

Tastes Great

Ready in minutes

Fully Cooked

Keep Frozen

NET WT: 11.25 OZ (320g)